Part 3 - Sound localisation

> **Note:** In "Sound localisation" mode, the "Export" button exports the comparison window with arrival-time and level difference. You can change the position of the sound source with the slider or directly in the simulation field.

The brain can determine the direction of a sound source partly because sound does not arrive at both ears in exactly the same way. Depending on the position of the sound source, differences arise in the arrival time and in the loudness of the signal at the left and right ear.

Task 1: Comparing two positions

First set the sound source to "front" (0 deg) and then clearly to one side, for example "right" (90 deg). Export the comparison window in both cases. Then compare the arrival times and level displays of both ears.

Sound source (front)
Insert here
Sound source (right)
Insert here

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Task 2: Investigating several positions

Now investigate at least four positions of the sound source. Enter the position of the sound source each time and add which ear hears the sound first, which ear receives the stronger level and roughly how large the differences are.

Position of the sound source	Which ear hears first?	Which ear hears louder?	Comment on the difference

Task 3: Explaining sound localisation

a) Use your observations to explain how the brain can infer the position of a sound source.

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b) Explain how the brain can distinguish a sound source in front (0 deg) from one behind (180 deg).

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